



UPP Mathematics

Mock Test 1

Question 1. Evaluate the following improper integral:

$$\int_0^1 \frac{\ln(x)}{\sqrt{x}} dx.$$

Question 2. Use Ratio Test to determine the convergence of the series:

$$\sum_{k=1}^{\infty} \frac{k^4 \cdot 4^k}{(2k)!}$$

Question 3. Determine the interval of convergence of the power series:

$$\sum_{k=2}^{\infty} \frac{3^k}{k^2} (x-1)^k$$

Question 4.

The line ℓ_1 passes through point A with coordinates $(3, -1, 4)$ and is parallel to the vector $\begin{bmatrix} 4 \\ -2 \\ 5 \end{bmatrix}$.

The plane H contains the point C with coordinates $(1, -3, 2)$ and the line ℓ_1 . Determine:

- (a) The standard equation of the plane H .
- (b) Symmetric equation of the line ℓ_2 containing the point C and is perpendicular to the plane H .

Question 5.

Construct the standard equation of the tangent plane and parametric equations for the normal line to the surface given by $e^x \cos(y) + e^y \sin(z) - \ln(y+1) - e = 0$ at the point $(1, 0, \pi)$.

Question 6.

Find and classify the critical points of the function $f(x, y) = x^2y + y^3 - 3x^2 - 3y^2 + 6$.